VG SERIES 3 , EPISODE 6 :

The Complete Hamiltonian and Its Reductions – From Network to Known Physics

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ABSTRACT

This episode presents the complete form of the Hamiltonian of the Primary Network, including all the terms necessary to describe node vibrations, couplings, spin, and torsion. It is shown that this structure reduces, in the appropriate limits, to Einstein's equations (general relativity) and to the Schrödinger equation (quantum mechanics). The evolution equation for the average coupling is also introduced, linking cosmic expansion to the relaxation of the Network. The episode includes a mathematical appendix for those who wish to verify the intermediate steps.

Keywords: complete Hamiltonian, Primary Network, general relativity, quantum mechanics, average coupling, Hubble Tension, VG Series 3.

1. INTRODUCTION – WHY WE NEED A COMPLETE HAMILTONIAN

In previous episodes, we used simplified forms of the Network Hamiltonian. These were sufficient to describe general phenomena—cosmology, Network structure, regime transitions. However, to connect RTS with mainstream physics, we need a complete form that includes all relevant terms and allows rigorous derivations.

This Hamiltonian is not a simple collection of terms. It is a unified structure from which both general relativity and quantum mechanics can be derived as limits of the same formalism.

2. THE COMPLETE HAMILTONIAN OF THE PRIMARY NETWORK

The complete form of the Hamiltonian is:

$$H = \sum_i (\frac{p_i^2}{2} + \frac{1}{2}\omega_i^2 q_i^2) + \sum_{i<j} K_{ij} q_i q_j + \sum_{i<j} \lambda_{ij} (\hat{S}i \cdot \hat{S}_j) q_i q_j + \sum i < j \gamma_{ij} (\hat{S}_i \times \hat{S}_j) \cdot q_i p_j$$

Meaning of the terms:

Vibrational term:

$$\frac{p_i^2}{2} + \frac{1}{2}\omega_i^2 q_i^2$$

Describes the individual vibration of each node. The frequency ω_i is determined by the spins of the node's links.

Scalar coupling term:

$$\sum_{i<j} K_{ij} q_i q_j$$

Describes the interaction between two nodes, independent of spin orientation. It is the source of gravity in the continuum limit.

Spin-spin coupling term:

$$\sum_{i<j} \left(\hat{S}_i \cdot \hat{S}_j\right) q_i q_j \, \lambda_{ij}$$

How spin alignment affects the interaction. If spins are aligned, the coupling is allowed; if anti-aligned, it is reduced.

Spin-orbit coupling term:

$$\sum_{i<j} \gamma_{ij} \left(\hat{S}_i \times \hat{S}_j\right) \cdot q_i p_j$$

Introduces rotation into the Network dynamics. It is the source of the torsion field.

3. REDUCTION TO EINSTEIN'S EQUATIONS (GENERAL RELATIVITY)

To obtain general relativity, we take the continuum limit: the number of nodes $N \to \infty$, the distance between nodes $a \to 0$, and the couplings become continuous functions of coordinates.

In this limit:

· The vibrational term becomes a scalar field.

· The scalar coupling term generates metric curvature.

· The spin terms become negligible for macroscopic phenomena.

The result is Einstein's equation:

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}$$

where:

· $G_{\mu\nu}$ is the Einstein tensor (curvature).

· Λ is the cosmological constant, which appears as the resid ual tension of the Network.

· $T_{\mu\nu}$ is the energy–momentum tensor (matter).

This reduction shows that gravity is an emergent effect of nodal resonance, not a fundamental force.

4. REDUCTION TO THE SCHRÖDINGER EQUATION (QUANTUM MECHANICS)

To obtain quantum mechanics, we take the weak-coupling limit: the interactions between nodes are small enough to be treated as perturbations.

In this limit:

· The Hamiltonian separates into a free part (individual vibrations) and an interaction part (couplings).

· The wave functions of the nodes become approximately independent.

· The dynamics are governed by the Schrödinger equation:

$$i\hbar\frac{\partial\psi}{\partial t} = H\psi$$

This reduction shows that quantum mechanics is a special case of nodal vibrations, in the regime where couplings are weak enough not to perturb individual coherence.

5. THE AVERAGE COUPLING EVOLUTION EQUATION (HUBBLE TENSION)

An essential result of RTS is the relation between cosmic expansion and the relaxation of the average coupling. This is formalized by the equation:

$$H(t) = -\frac{1}{\alpha}\cdot\frac{\dot{J}med(t)}{Jmed(t)}$$

where:

· $H(t)$is the Hubble constant at cosmic time t.

· $J_{med}(t)$ is the average coupling of the Network at time t.

· $\dot{J}_{med}(t)$ is the rate of change of the average coupling.

· α is a dimensionless constant relating geometry to Network dynamics.

Physical interpretation:

The accelerated expansion of the universe is the relaxation of the Network's average coupling. When the coupling decreases rapidly, expansion accelerates. When it stabilizes, expansion slows down.

This equation provides a natural explanation for the Hubble Tension—the discrepancy between local and cosmological measurements of the Hubble constant. In RTS, this tension is a signature of Network relaxation, not a measurement error or an exotic particle.

6. CONNECTION TO THE GRAND EQUATION (RTS18)

The complete Hamiltonian integrates into the Grand Equation from RTS18:

$$H_{\text{total}} = H + \mathcal{L}_{vivo}$$

where $\mathcal{L}_{vivo}$ is the term describing intention, consciousness, and self-creation of the Network.

To include spin in this structure, we can extend $\mathcal{L}_{vivo}$ as:

$$\mathcal{L}_{vivo} = \frac{1}{2}\dot{\phi}^2 - V(\phi) + \lambda \cdot \phi \cdot \psi + \mu \cdot \phi \cdot \mathcal{I} + \nu \cdot \hat{S} \cdot \phi$$

where the last term links spin to the state of the Network, opening the way to a unified description of matter, energy, and consciousness.

7. CONCLUSION – A MATHEMATICAL UNIFICATION

VG6 has shown that:

· The complete Hamiltonian of the Primary Network can be written in a rigorous form.

· It reduces to general relativity in the continuum limit.

· It reduces to quantum mechanics in the weak-coupling limit.

· It provides a natural explanation for the Hubble Tension.

· It integrates into the Grand Equation, opening the way to a unified theory of matter, energy, and consciousness.

These results are not merely formal. They show that RTS is a theory that can be verified, extended, and discussed in scientific terms.

MATHEMATICAL APPENDIX – FOR THE ATTENTIVE READER

A.1. Derivation of the average coupling evolution equation

Starting from the complete Hamiltonian, one can show that the average coupling $J_{med}(t)$ satisfies:

$$\dot{J}med = -\alpha H(t) Jmed$$

which, by integration, leads to:

$$J_{med}(t) = J_{med}(t_0) \cdot \exp\left(-\alpha \int_{t_0}^{t} H(t')dt'\right)$$

This shows that the relaxation of the average coupling is directly linked to cosmic expansion.

A.2. Reduction to Einstein's equation

In the continuum limit, the spin terms become negligible for macroscopic phenomena. The Hamiltonian reduces to:

$$H_{\text{cont}} = \int d^3x\left(\frac{\pi^2}{2} + \frac{1}{2}(\nabla\phi)^2 + V(\phi)\right)$$

from which the field equations of general relativity are obtained.

A.3. Reduction to the Schrödinger equation

In the weak-coupling limit, the Hamiltonian separates into a free part and an interaction part. The wave functions of the nodes satisfy:

$$i\hbar\frac{\partial\psi_i}{\partial t} = \left(-\frac{\hbar^2}{2}\nabla^2 + V_i\right)\psi_i$$

which is exactly the Schrödinger equation for a particle in the potential V_i.